DESIGN THINKING AND INNOVATION(DTI)(CSET210)

BENNETT UNIVERSITY
TIMES OF INDIA GROUP

QUEUE MANAGEMENT SYSTEM

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Queue Management System

Web Application for Restaurant Management

OVERVIEW

Introduction

What is this Project?

This project is a web-based restaurant management system that allows customers to book tables, order food, and track deliveries—improving queue handling and overall customer experience.

The system integrates queue management, online ordering, and real-time delivery tracking into a single unified digital platform for restaurants.

PROBLEM

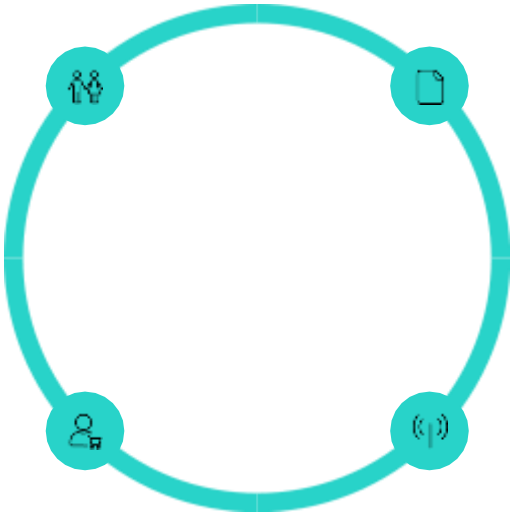
Problem Statement

Long Waiting Queues

Customers face excessive wait times at restaurants, especially during peak hours, leading to frustration and loss of business.

Poor Customer Experience

During peak hours, the combination of queues, delays, and lack of information results in a poor and frustrating dining experience.



Manual Booking Inefficiency

Traditional manual table booking processes are slow, error-prone, and unable to handle high volumes of reservations effectively.

Lack of Real-Time Updates

Customers receive no live status on table availability, order progress, or estimated delivery times once an order is placed.

GOALS

Objectives

These objectives drive the entire system design, ensuring every feature delivers measurable value to both customers and restaurant operators.

Automate Queue Management

Eliminate manual queue tracking and reduce customer wait times through intelligent digital queue handling.

Enable Online Table Booking

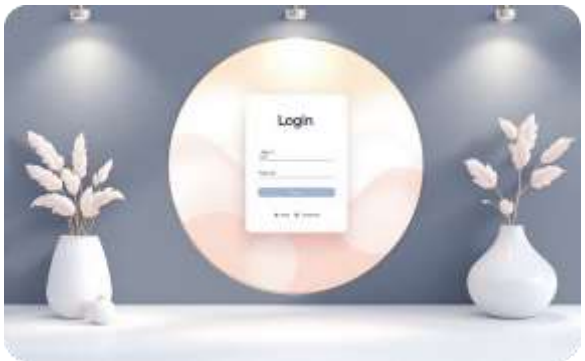
Allow customers to reserve tables from anywhere, at any time, based on real-time availability.

Real-Time Order Tracking

Provide live updates on order status and delivery progress so customers are always informed.

ARCHITECTURE

System Overview



User Registration & Login

Customers register and securely log in to access the full suite of features including table booking and food ordering.

Table Booking

Customers book tables based on live availability. The system automatically manages the queue and notifies when tables are ready.

Order Food & Track Delivery

Customers browse the menu, place orders, and track real-time delivery status via an integrated map. All data is stored in the databas

PROCESS

Working Process

End-to-End Customer Journey

From registration to delivery tracking —every step is seamlessly handled by the system.

01

Register & Log In

User creates an account and logs in securely using OTP phone number verification via Twilio Authy.

02

Book a Table

Customer selects a table based on availability; queue management begins automatically in the system backend.

03

Order Food

Customer browses the digital menu, adds items to cart, and places their food order online.

04

Order Processing

The system processes the order and updates kitchen and delivery status in real time.

FEATURES

Key Features

Table Reservation System

Real-time table availability management with automated queue handling to minimize customer wait times.

Online Food Ordering

Full-featured digital menu with cart system, allowing customers to order food from anywhere at any time.

Delivery Tracking via Maps

Live delivery tracking powered by Google Maps API so customers always know where their order is.

Phone OTP Verification

Secure phone number verification using Twilio Authy to authenticate users during registration and login.

User Authentication System

Robust user authentication ensuring only verified customers can access the booking and ordering features.

STACK

Technologies Used

Frontend

HTML, CSS, JavaScript — Responsive and interactive user interface for seamless customer experience across devices.

Backend

Flask (Python) — Lightweight and powerful web framework handling server-side logic, routing, and API integrations.

Database

SQLite — Embedded relational database storing all user, booking, order, and delivery data reliably.

APIs

Twilio Authy for OTP verification and Google Maps API for real-time delivery tracking integration.

MODULES

System Modules

User Registration & Login

Handles new user sign-up, secure login, and OTP-based phone verification for authenticated access.

Order Management & Delivery Tracking

Processes placed orders, updates statuses, and tracks delivery progress in real time via Google Maps API.

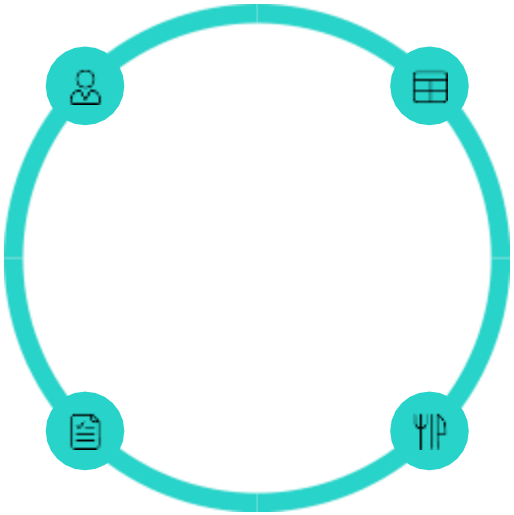


Table Reservation Module

Manages real-time table availability, booking confirmations, and queue ordering for dine-in customers.

Menu & Cart System

Displays the restaurant menu, allows item selection, and manages the customer's cart before checkout.

WRAP-UP

Advantages, Limitations & Future Scope

✅ Advantages

- Reduces waiting time through automated queue management
- Enables convenient online table booking
- Improves customer convenience with real-time tracking
- Increases restaurant e□iciency and profitability

🔮 Future Enhancements

- AI-based wait time prediction
- Mobile application integration
- Online payment gateway
- Smart queue notifications

⚠️ Limitations

- Requires a stable internet connection
- Dependency on third-party APIs (Twilio, Google Maps)
- Initial setup complexity for restaurant staff
- Requires ongoing maintenance and updates

🔚 Conclusion

The system provides a complete digital solution for restaurant management —integrating queue management with online ordering and real-time delivery tracking to enhance both customer experience and overall business performance.

Reserve a Table

Fill out the form below to make a reservation.

Full Name *

Full Name

Email *

customer@gmail.com

Phone Number

Phone Number

Reservation Date *

04/13/2026

Reservation Time *

12:30 PM

Table *

Number of People *

Number of People

Special Requests

Special Requests

Reserve Now

BookMyTable

Home Menu About Contact Account

Reserve Your Table at BookMyTable

Discover a delightful dining experience at BookMyTable. Book your table seamlessly with our easy-to-use platform. Enjoy exquisite dishes, impeccable service, and create memorable moments.

Book your table now

ABOUT US

We offer premier services where excellence meets affordability.

TABLE RESERVATION

Effortlessly reserve tables for your dining experience, ensuring you have the perfect spot.

EVENT RESERVATIONS

Book special events or private dining experiences for groups, celebrations, or business meetings.

ORDER HISTORY

Access your order history to view past reservations, reorder favorite meals, and manage dining preferences.